

ROLL NO :- 04

Assignment :- Organic Chemistry 2nd Semester.

Solvent effect:-

Change in the reactivity or properties of a solute caused by presence of a solvent.

Solvents can impact various aspects of solute behaviour including:-

Reactivity :- Solvents can alter the rate or outcome of a chemical reaction.

Solubility :- Solvents can effect the ability of a solute to dissolve.

Stability :- Solvent can affect the stability of solute, potentially leading to degradation or decomposition.

Spectroscopic properties:- Solvents can shift or broaden spectroscopic signals (eg NMR, IR)

Chemical equilibrium:- Solvents can alter the equilibrium position of a chemical reaction.

Solvent can be attributed to various factors:-

Polarity:-

Solvents can be polar or non-polar, influencing the solute's behaviour.

Dielectric constant:-

The Solvents ability to dissolve ions and polar

molecules.

Hydrogen bonding:-

Solvents can form hydrogen bonds with solute impacting its behaviour.

Steric effect:-

Solvents can influence the spatial arrangement of molecules.

Solvent effect on the acidity of alcohols can be explained in following ways:-

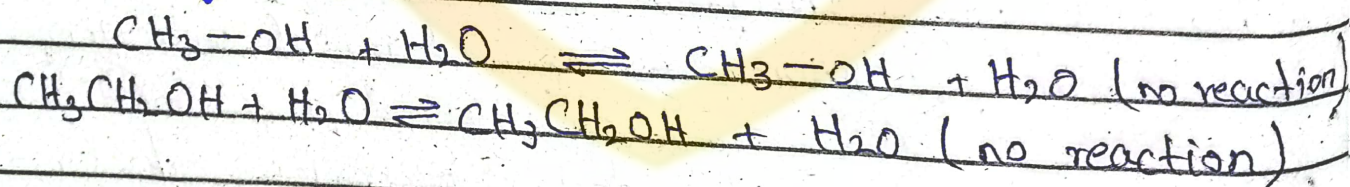
i- **In aqueous solution:-** Alcohols are neutral and do not change pH of solution.

Example:-

for example.

Methanol and ethanol are neutral in water, do not change the pH.

equation:-



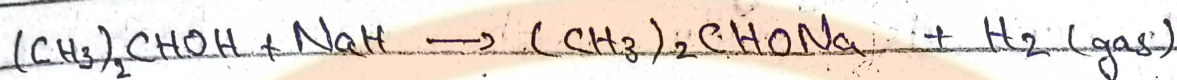
ii **In organic solvents** the acidity of alcohols can be enhanced, allowing them to be fully deprotonated by strong bases like Sodium metal.

Sodium metal (Na) can deprotonate ethanol

in tetrahydrofuran (THF) to form a sodium ethoxide.



Sodium hydride can deprotonate ethanol isopropyl alcohol in dimethyl sulfoxide (DMSO) to form a sodium isopropoxide.

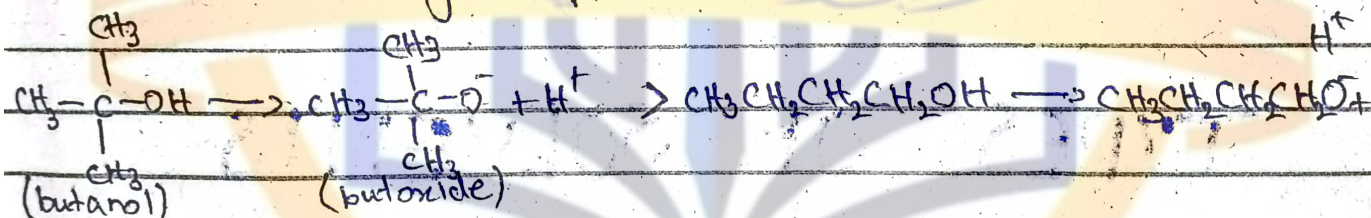


iii - The acidity of alcohols in gas phase is

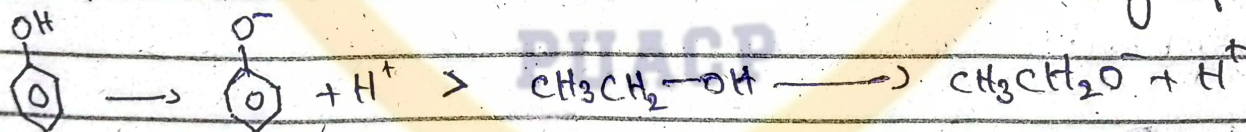
reversed compared to in solution, with tertiary alcohols being more acidic than primary alcohols

Example:-

1- tertiary butanol is more acidic than primary butanol in gas phase.



2- Phenol is more acidic than ethanol in gas phase



iv Solvation effect

play a crucial role in determining the acidity of alcohols with larger alkoxide ions being less well solvated.

Lithium alkoxide are more stable in solvents like

THF or diethyl ether due to coordination with metal ions.



- 2- Potassium tert-butoxide ($\text{K}^+\text{O}^-\text{t-Bu}$) is more stable in DMSO than in water due to more solvation.



✓ The size of conjugate base of alcohols influences its acidity with larger base being more stable being less acidic.

- 1- Larger alkoxide ions like potassium tert-butoxide are less acidic than smaller ones like potassium ethoxide.
- 2- Sodium phenoxide ($\text{Na}^+\text{O}^-\text{C}_6\text{H}_5$) is less acidic than sodium methoxide due to larger size of sodium phenoxide ion.

TYPES OF SOLVENTS

Solvents are broadly classified on bases of polarity and chemical nature.

Based on polarity:-

Polar solvents

Non-polar solvents

• Polar solvents

have high dielectric constant and dissolve ionic and polar compounds.

• Non-polar solvents

are better at dissolving non-polar compounds.

Based on proticity:-

• **Protic solvents** can donate a proton and often participate in hydrogen bonding. They can stabilize a conjugate base due to hydrogen bonding which results in increasing acidity of alcohols. ethanol is more acidic in water as compared to be in non-polar solvent.

• **Aprotic solvents** can not donate a proton they stabilize ion through dipole interaction they increase the stability of alcohol by stabilizing alkoxide ion.

ACIDITY OF PRIMARY, SECONDARY AND TERTIARY ALCOHOLS

Their acidity depends upon the stability of alkoxide ion more stable the ion is the more acidic the alcohol will be.

Primary alcohols:-

Alkoxide ion derived from a primary alcohol is least sterically hindered and can be better stabilized by solvent molecule makes them more acidic.

Secondary alcohols:-

Alkoxide ion is more sterically hindered than primary reducing the stabilization by solvent consequently they are less acidic.

Tertiary alcohols:-

The alkoxide ions are highly hindered, making it difficult for alcohols to stabilize making them least acidic.

So the order of acidity is:-

Primary > Secondary > tertiary

CONCLUSION:-

Solvent effect as well as the nature of alcohol plays a very important role in determining acidity of alcohols which depends upon the stabilization of alkoxide ion in a solution.